

CURRENT AFFAIRS



**S & T HEALTH
ENVIRONMENT**



36th WEEK

MODULE - 2

SAHARA FOREST PROJECT!

DO WE HAVE ENOUGH TO PROVIDE US ALL FOREVER ???

WHAT DO WE HAVE IN ABANDUNCE ?

SALT WATER FROM OCEANS
CARBON DIOXIDE
SUNLIGHT
WIND



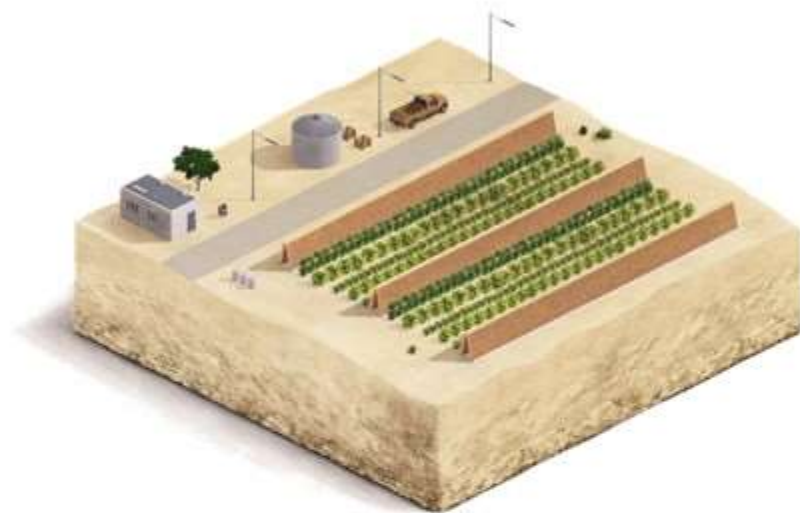
WHAT DO WE NEED AT PRESENT ?

FRESH WATER
FOOD FOR ALL
CLEANER ENERGY
GREENERY IN DESERT REGIONS



GREENERY IN HOT DESERTS! A NEAR POSSIBILITY...

DESIGN LAYOUT



ACTUAL SITE





SUSTAINABLE USE OF DESERT & SEA... SAHARA FOREST PROJECT



Plants growing in desert soils in Qatar, as a part of the pilot Sahara desert Project

- Earth is 70 % covered with salty oceans but the fresh water is depleting day by day.
- The human population is increasing and we need more food to feed the extra mouths.
- World is covered with vast areas of deserts which are still uninhabited.
- Thus sustainable development is required to combine what the Earth already we have in abundance into what we need!
- This is why a novel project started in the Qatar deserts and now initiated in Jordan.



THE LAYOUT OF SAHARA FOREST PROJECT





SUSTAINABLE USE OF RESOURCES... SAHARA FOREST PROJECT



Under the patronage of His Majesty King Abdullah II of Jordan and His Royal Highness Crown Prince Haakon of Norway the Sahara Forest Project Launch Station in Jordan was inaugurated on Thursday September 7th 2017.

- A full scale 3 hectare SFP is established in Aqaba, Jordan.
- The launch station has a saltwater cooled greenhouse with solar power technologies and facilities for outdoor cultivation and revegetation.
- The Project is funded by European Union and Norwegian Government.
- Joakim Hauge is the CEO of the SFP.

DESERT AREA WITH SEA WATER FACILITY- AQABA JORDAN



JORDAN

AQABA





KEY HIGHLIGHTS OF SAHARA FOREST PROJECT



- Inaugurated in Aqaba Jordan on September 7, 2017
- In the first stage, the project aims to produce upto 130 tonnes of organic vegetables per year from an area of 4 football pitches.
- The project will use solar panels to provide and include outdoor planting space, two saltwater cooled greenhouses unit and salt ponds for salt production.
- The project will be expanded from 3 hectares to around 200 hectares of desert.

CORE TECHNOLOGIES USED IN SAHARA FOREST PROJECT

1

SALTWATER COOLED GREENHOUSE- the evaporation of saltwater to create cooling effect for vegetation growth in greenhouse and obtaining distilled fresh water.



CORE TECHNOLOGIES USED IN SAHARA FOREST PROJECT

- 2** **CONCENTRATED SOLAR POWER-** to generate huge amounts of solar energy by converging large area of sun's rays using mirrors or lenses.



CORE TECHNOLOGIES USED IN SAHARA FOREST PROJECT

3

DESERT REVEGETATION- by using evaporative hedges, the surrounding area of the green house can be cooled with salt water. This can provide warm, humid condition out in the desert to grow crops. Halophytes like date palm, barley etc. can be grown in commercial scale.





SAHARA DESERT PROJECT IN JORDAN BOTTOM LINE IDEOLOGIES...



Jaokim Hauge

The Sahara Forest Project is founded on the idea that deserts once again can be green.

This can be realized through activities with a triple bottom-line Ideologies.

- 1) The activities need to be good for people,
- 2) It need to good for the environment,
- 3) It needs to be good for business, as quoted by Mr. Hauge



WHAT ABOUT THE ECONOMIC DEVELOPMENTS FROM SAHARA FOREST PROJECT ?



SALT EXTRACTION



ALGAE PRODUCTION



MARICULTURE FROM SEA WATER



HALOPHYTE CULTIVATION

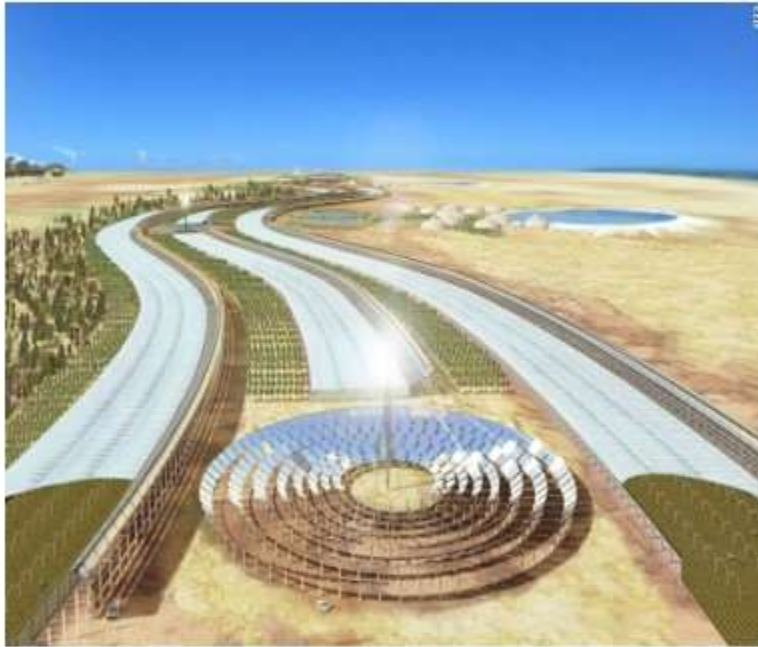


BIOFUEL PRODUCTION





SAHARA FOREST PROJECT REVIVING DESERTS FOR MANKIND



Artist's impression of SFP

- Deliver useful resources to us like food fresh water and clean energy.
- It will generate many products with high economic value.
- It will create jobs for the local communities both skilled and unskilled.
- All this will revegetate the desert lands which will boost the ecosystem.
- It will create cleaner environment with economic growth in the barren deserts of the World.

Sahara Forest Project as said is truly 'Too Good to be True' !!!

QUESTION - 1

Look at the following regarding the Sahara Forest Project :

- 1) It was inaugurated in Aqaba, Jordan in September 2017
- 2) It is a ambitious project by Norway and European Union.
- 3) It uses salt water cooled green houses and solar power
- 4) It is launched with an aim to revegetate the uninhabited desert

Select the correct option using the codes below:

- | | |
|-------------------|--------------------|
| a) 1,2 and 4 only | b) 1,2 and 3 only |
| c) 2,3 and 4 only | d) All 1,2,3 and 4 |

QUESTION - 2

Saltwater cooled greenhouse was in news recently. Which of these statements best explains it ?

- 1) It is used in the deserts of Jordan for revegetation.
- 2) It provides a conducive atmosphere to grow crops in desert.
- 3) It evaporates the sea water to provide adequate humidity to the crops grown inside it.
- 4) It is a technology used in the Sahara Forest Project

Select the correct option using the codes below:

- | | |
|-------------------|--------------------|
| a) 1,2 and 4 only | b) 1,2 and 3 only |
| c) 2,3 and 4 only | d) All 1,2,3 and 4 |

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Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

Visit us at:

www.learningspacedigital.com

OUR TEAM

G. V. Rao

D. Sunil Kumar

V. Dayakar James

Y. Sri Poojita

Amrita Naidu

G. Ravi Babu

D. Joji

K. Victor Babu

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